M0: BERT-Tiny CV–job fit classifier (L=2, H=128, A=2, 256 tokens)

Parameters
MACs per pair (256 tokens)

Input

CV text + job description text

WordPiece tokenizer (uncased, 30,522-token vocabulary)

[CLS] CV tokens [SEP] job tokens [SEP] → 256 ids

longest-first truncation · segment ids 0/1 · padding mask

0

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Embeddings

word 30,522×128 + position 512×128 + segment 2×128

→ LayerNorm, dropout → 256 × 128 vectors

3,972,864 (90.6% of all)

≈0 (table look-ups)

Transformer encoder layer 1

Multi-head self-attention, 2 heads × 64:

Q, K, V, O = four Linear 128→128, softmax($QK^T/\sqrt{64}$)·V

Add & LayerNorm

Feed-forward: Linear 128→512 · GELU · Linear 512→128

Add & LayerNorm → 256 × 128

198,272

67.1 M (projections 16.8 M, attention 16.8 M, FFN 33.6 M)

Transformer encoder layer 2

Multi-head self-attention, 2 heads × 64:

Q, K, V, O = four Linear 128→128, softmax($QK^T/\sqrt{64}$)·V

Add & LayerNorm

Feed-forward: Linear 128→512 · GELU · Linear 512→128

Add & LayerNorm → 256 × 128

198,272

67.1 M (projections 16.8 M, attention 16.8 M, FFN 33.6 M)

Pooler + classifier head

[CLS] vector → Linear 128→128, tanh (pooler)

dropout → Linear 128→3 (new, randomly initialised)

16,512 + 387

16.4 K + 0.4 K

Output

softmax over No Fit · Potential Fit · Good Fit

M1 change

Teacher BERT-base (fine-tuned, training only) gives soft targets

loss = 0.5·CE + 0.5·T^2·KL

M2 change (post-training):

dashed purple boxes are quantized
• all Linear layers → dynamic INT8
• embeddings → 8-bit weight-only
• LayerNorm stays FP32

Total: 4,386,307 parameters
≈134.2 M MACs per pair